

# Yejun Yeo

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## Education

- University of Waterloo B.S.C in Computer Science Co-op Sep 2024 – May 2029

## Work Experience

*Machine Learning Engineer at City of Edmonton*

*Jan 2026 to Aug 2026*

- Built a Machine Learning model that predicts wildfires in Edmonton using R, chosen for its geospatial libraries, trained on weather and land data pulled from PostgreSQL.
- Ported the R/Shiny wildfire prediction model to FastAPI + React/TypeScript, including hand-porting R fire prediction libraries to Python, achieving a 6x faster load time (2 min to 20 s).
- Mentored by Senior Architects to design production pipeline using Docker, GitLab CI/CD, and Google cloud (GKE), implementing features based on their reviews.
- Collaborated with 3 senior engineers through Git feature branches and merge requests, and iterated on the app based on bi-weekly stakeholder feedback.

## Projects

*Brain Tumour MRI Classifier*

- Built a CNN in PyTorch and Python to classify brain tumour MRI scans from a Kaggle dataset into four categories
- Pre-processed images with OpenCV and NumPy by resizing, grayscaling, and normalizing to give the model consistent input
- Tuned the PyTorch architecture with ReLU, max pooling, and softmax layers, and tracked accuracy and loss curves in Matplotlib to catch overfitting

*Machine Learning Based PRNG Research Paper*

- Authored a 4,000-word research paper comparing the predictability of PRNGs (Pseudorandom Number Generators) used in Python and Java.
- Implemented linear regression with gradient descent in Python to evaluate error margins across 400×25 datasets of pseudo-random numbers.
- Analysed results showing Python's PRNG (Mersenne Twister) is slightly more predictable than Java's LCG (Linear Congruential Generator)